



#79 PAPER 110 - ALMOST PERFECT SCALE - VALIDITY OF A PERFECTIONISM SCALE ON A ROMANIAN UNIVERSITY SAMPLE

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Abstract

The aim of the study was to investigate the psychometric properties of the Almost Perfect Scale-Revised. We used both exploratory and confirmatory factor analysis in order to study the construct validity of the scale. The participants were undergraduate psychology students. The reliability estimates for the three APS-R subscale scores showed a good internal consistency. The exploratory factor analysis confirmed a two factor structure. Using the confirmatory factor analysis, the two-factor model approached an acceptable fit. Correlations between APS-R subscales and grade-point average and other perfectionism measures provided convergent and divergent validity support.

Keywords: *academic adjustment; academic performances; perfectionisms; factor analysis.*

1. INTRODUCTION

Perfectionism has been extensively studied in the recent years. While the initial research studied perfectionism as a unidimensional concept (Burns, 1980), nowadays the perfectionism is studied as a multidimensional concept, being assessed with different instruments, such as: The Multidimensional Perfectionism Scale (Hewitt & Flett, 1991), The Positive and Negative Perfectionism Scale (Terry-Short, Glynn Owens, Slade, & Dewey, 1995), The Adaptive/Maladaptive Perfectionism Scale (Fong & Yuen, 2011), The Almost Perfect Scale-Revised (Slaney, Mobley, Rice, Trippi, & Ashby, 2001). The most widely used seems to be the Almost Perfect Scale-Revised (APS-R).

The Almost Perfect Scale-Revised is a 23-item measuring three scales (High Standards, Discrepancy, and Order), designed to measure the multidimensional construct of perfectionism. The High Standards subscale (7 items) measures high personal standards for performance and achievement. The Discrepancy scale (12 items) measures respondents' perceptions of themselves as failing to meet their personal standards for performance. The Order subscale (4 items) measures a preference for neatness and order. The authors of the instruments argued that perfectionism is not an inherently maladaptive personality dimension; instead, it is multidimensional, with some aspects clearly maladaptive but others clearly adaptive (Slaney, Rice, & Ashby, 2002). Previous research showed that Discrepancy scores were associated with depression, anxiety, interpersonal problems, life (dis)satisfaction, and acculturative stress; High Standards scores were associated with positive indicators of well-being, such as self-esteem or academic performance (Rice, Richardson, & Tueller, 2014).

2. METHOD

2.1. Aim of the study

The aim of the study was to investigate the psychometric properties of the Almost Perfect Scale-Revised (APS-R, Slaney, Mobley, Rice, Trippi, & Ashby, 1999). The construct validity of the scale was investigated.

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2.2. Participants

The study was conducted on an adolescent sample including students from Transilvania University of Brasov (N = 200), 159 female and 41 male, aged between 19 and 25, with a mean age of 21 years. All the participants were undergraduate psychology students. The participants gave their informed consent to participate in the study. The questionnaires were completed during class time, were anonymous and no compensation was offered.

2.3. Instruments

The Almost Perfect Scale–Revised (APS-R, Slaney et al., 2001) was used. The 23 items are measured on a 7-point, Likert-type scale ranging from 1 (strongly disagree) to 7 (strongly agree). Higher scores correspond to more perfectionism. The three dimensions: High standards, Discrepancy and Order cover two dimensions: adaptive and maladaptive dimensions of the perfectionism. High standards and order represent forms of adaptive perfectionism, while order represents a maladaptive perfectionism.

The Academic Adjustment Questionnaire (AAQ, Clinciu & Cazan, 2014) was also used. AAQ is a self-report instrument scored with 0 and 1, designed to assess the student's adjustment to academic learning. It comprises 24 items, from which 14 assess Neuroticism associated to academic learning and 10 items assess Procrastination.

Academic performance was also taken into consideration, the participants declared their grade point average at the end of the previous semester.

3. RESULTS

The construct validity of the scale was tested through exploratory and confirmatory factor analysis. An initial exploratory factor analysis was used, in order to test the factorial structure of APS-R. The Kaiser-Meyer-Olkin (KMO) (KMO values > .6) and Bartlett's test of sphericity (Chi-square [χ^2] significant at $p < .05$) indicated that the data was suitable for factor analysis (KMO = .89; Bartlett's test of sphericity (χ^2 (253) = 2575.65, $p < .000$)). The initial exploratory factor analysis confirmed the original structure of the instrument, with three scales but the third factor was problematic, given the fact that only two items loaded into this dimension. A second factor analysis revealed a two-factor solution, highlighting accurately the two dimensions, adaptive perfectionism (including all the items from the Standards and Order scales excepting item 22) and maladaptive perfectionism (including all the items of the Discrepancy scale and item 22). Eigenvalues and an evaluation of the scree plot indicated a two factor solution explaining 28.47% and 21.74% of the variance respectively (Tabachnick & Fidell, 2001). The item 22 seems to be problematic, loading weakly on another dimension that the theoretical structure of the instrument. We computed internal consistencies with and without Item 22 for each subscale to assess whether this may affect measure reliability, with Cronbach Alpha (α) of above .70 indicating a reliable measure (Tabachnick & Fidell, 2001): Adaptive perfectionism with item 22, $\alpha = .86$, Adaptive perfectionism without item 22, $\alpha = .85$, Maladaptive perfectionism with item 22, $\alpha = .91$, Maladaptive perfectionism without item 22, $\alpha = .90$. Although the presence of item 22 do not improve significantly the scale reliability, it was however retained in the confirmatory factor analysis for a more deeper analysis, in order to evaluate whether it may be retained for one scale or its removal would best fit the model (Table 1).

Table 1. Exploratory factor analysis for the APS-R

	CommunalitiesTotal	
	F1	F2
APSR20	.81	.67
APSR21	.80	.65
APSR13	.78	.22
APSR11	.78	.63
APSR17	.78	.62
APSR16	.76	.58
APSR19	.74	.55
APSR23	.70	.50
APSR6	.67	.49
APSR15	.67	.48
APSR9	.59	.35
APSR3	.50	.27

APSR22	.29	.09
APSR10	.83	.70
APSR7	.78	.65
APSR8	.77	.62
APSR12	.70	.49
APSR2	.70	.49
APSR1	.67	.46
APSR4	.65	.43
APSR18	.65	.42
APSR14	.60	.36
APSR5	.52	.32
N of items	13	10
Eigenvalues	6.81	4.73
% explained variance	28.47	21.74
Alfa Cronbach	.91	.88

Extraction Method: Principal Component Analysis.
 Rotation Method: Varimax with Kaiser Normalization.
 Communalities lower than .25 were omitted

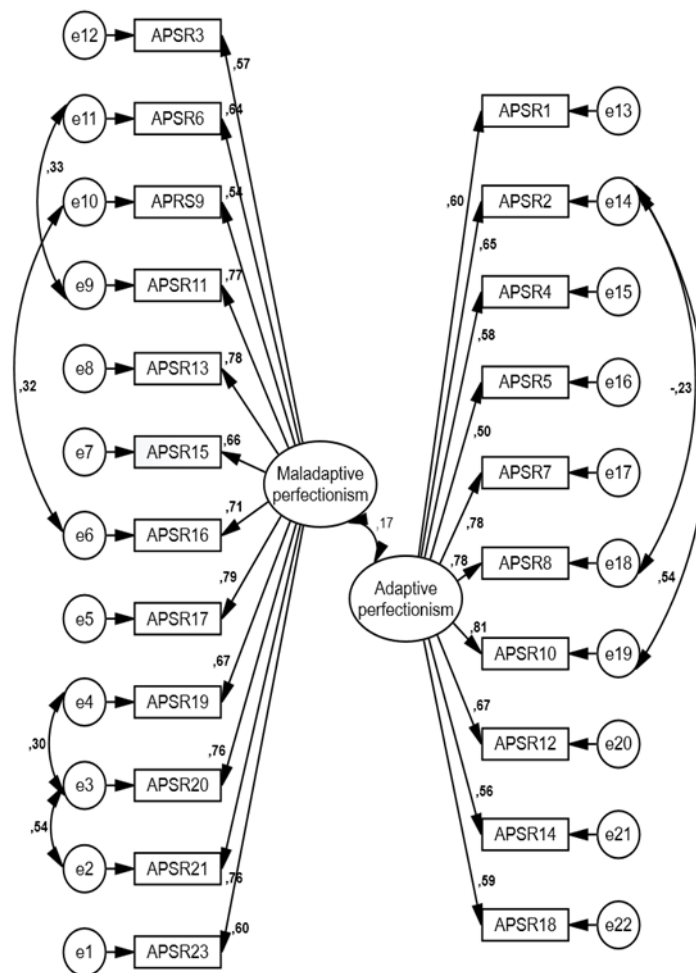


Fig. 1. Confirmatory factor analysis for the APS-R (standardized estimates)

We used confirmatory factor analysis in order to test the construct validity of the scale. Assessment of normality and outliers suggests the following results: z-statistics of 7.59 indicates a slightly non-normality of the sample. Although, there are no multivariate outliers, Mahalanobis distance showed minimal evidence of multivariate outliers. We computed several models: a first model with three factors (according to the initial structure of the instrument) with and without correlated errors, two models with two factors (as indicated by the

exploratory factor analysis) with and without correlated errors, a model with item 22 included in the maladaptive dimension, and a last model without item 22 (Table 2). The correlations between the three scales (Order, Discrepancy and Standards) showed weak correlations between Standards and Discrepancy ($r = .18$, $p = .01$), no significant correlation between Order and Discrepancy ($r = .12$, $p = .08$) and a high correlation between Order and Standards ($r = .66$, $p < .001$).

The modification indices analysis also suggested evidence of misspecification associated with the pairing of several errors terms associated with items 11 and 6, 9 and 16, 19 and 20, 20 and 21, 2 and 8, 2 and 10 (Figure 1). The model fit was better for the model with correlated factors. The model fit did not improve significantly, thus we decided that the last model (two factor with correlated errors without item 22) has the best fit indicators and it is preferable being the most parsimonious model. Reviewing the unstandardized estimates, all are statistically significant, given the critical ration (CR) values higher than 1.96.

Table 2. Goodness-of-fit measures for the tested models – First order CFA for the APS-R

Model	$\chi^2(df)$	GFI	CFI	AIC	RMSEA (90%CI)
1. Three factor model	689.2 (228)	.768	.811	785.2	.101 (.092-.109)
2. Three factor model with correlated errors	560.14 (224)	.806	.862	664.14	.087 (.78-.096)
3. Two factor model	770.16 (230)	.740	.778	862.16	.109 (.100-.117)
4. Two factor model with correlated errors	659.6 (227)	.767	.823	.757.6	.098 (.089-.107)
5. Two factor model with correlated errors (item 22 loaded into maladaptive factor)	.646.51 (227)	.771	.828	744.51	.096 (.088-.105)
6. Two factor model with correlated errors without item 22	501.55 (203)	.814	.875	601.55	.070 (.068-.093)

Note. GFI: Goodness-of-Fit Index, CFI: Comparative Fit Index, AIC: Akaike Information Criterion, RMSEA: Root Mean Square Error of Approximation, 90% CI: 90% confidence interval for RMSEA.

The Pearson correlation coefficients showed a positive and significant association between adaptive perfectionism and academic performances. While adaptive perfectionism did not correlate with academic maladjustment, maladaptive perfectionism highly correlated with academic maladjustment. Both adaptive and maladaptive perfectionism correlated negatively, respectively positively with procrastination. Maladaptive perfectionism highly correlated with academic neuroticism (Table 3).

Table 3. Pearson correlation coefficients between perfectionism and academic (mal) adjustment

	Neuroticism	Procrastination	Academic maladjustment	Academic performance
Adaptive perfectionism	.033	-.196**	-.077	.240**
Maladaptive perfectionism	.593**	.213**	.510**	-.070

Note: N = 200, ** $p < .001$

4. CONCLUSIONS AND DISCUSSION

In this study, we examined the psychometric properties of the Almost Perfect Scale-Revised (APS-R; Slaney et al., 2001) in a sample of university students. The results showed that the APS-R has good psychometric properties in the Romanian context. Although the authors recommended the use of three scales, the factor analysis showed that the two-factor solution has a better fit and it is more efficient. We did not compute age and gender differences given the high homogeneity of the sample and the small number of boys. The low correlation between adaptive and maladaptive perfectionism was consistent with other research suggesting minimal shared variance between the two dimensions (Rice, Ashby, & Gilman, 2011). Although the literature in this field confirmed the three-factor solution of the scale (Diamantopoulou & Platsidou, 2014; Rice et al., 2014; Vandiver & Worrell, 2014), our study revealed a different structure, which did not differentiate between Order and Standards. On the other hand, although the Order showed good factorial structure and internal consistency in previous studies, some authors argued that it has not yielded much empirical predictive benefit in understanding perfectionism (Rice et al., 2014; Stoeber and Otto, 2006). The high correlation between maladaptive perfectionism and neuroticism was also confirmed in a previous study, showing that neuroticism could be a relevant indicator of the maladaptive perfectionism (Rice, Ashby, & Slaney, 2007).

Further research would focus also on these aspects and on the measurement invariance of the APS-R across distinct contexts. Pearson correlations indicated that adaptive perfectionism is positively related to grade-point average, whereas maladaptive perfectionism has a negative relationship with academic performance. Similar results regarding the adaptive perfectionism were previously reported in the literature (Vandiver & Worrell, 2014). Convergent and discriminant validity must also be checked. Other studies showed that APS-R scores highly correlated with other perfectionism measures, such as The Frost Multidimensional Perfectionism Scale and the Performance Perfectionism Scale (Rice et al., 2014). Emotion regulation was used as an additional criterion indicator, showing that maladaptive perfectionists had emotion regulation problems (Rice et al., 2014). Despite the promising results, the generalizability of the findings are limited to the university students. In

conclusion, this study provided preliminary support for the APS-R (Slaney et al., 2001) as a measure of perfectionism.

5. REFERENCES

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